

AWS-VPC -02

1. Create one VPC, with 1 public subnet and 1 private subnet.

- Created a VPC named MYPVC with CIDR block (e.g., 10.0.0.0/16).

	vpc-09bec8abbcd98cb749	Available	-	-	On
☒ MyVPC	vpc-01e8ea44337907063	Available	-	-	Off

- Configured one public subnet (e.g., 10.0.1.0/24) with Internet Gateway for external access.
- Configured one private subnet (e.g., 10.0.2.0/24) without direct internet access.
- Associated route tables:
- Public subnet → 0.0.0.0/0 → IGW
- Private subnet → internal/local routing only.
- Ensured proper subnet association and security group configuration.

Subnets (4) [Info](#) Last updated about 2 hours ago

☐	Name	Subnet ID	State	VPC
☐	Private-Subnet-1	subnet-08b1596d62f03f97e	Available	vpc-01e8ea44337907063 MyV...
☐	Public-Subnet-1	subnet-0ba5a9aa281715d09	Available	vpc-01e8ea44337907063 MyV...

2. Enable VPC peering for cross-region.

- Created VPCs in **different regions** (e.g., Ohio & N. Virginia).
- Established a **VPC peering connection** between the two regions.
- Accepted the peering request in the acceptor region.
- Updated route tables in both VPCs to enable communication using **peering connection ID**.
- Verified connectivity using **private IP communication between EC2 instances**.

pcx-0f73c41106c81555b

Pending acceptance
You can accept or reject this peering connection request using the 'Actions' menu. You have until Thursday, 23 April 2026 at 15:34:37 GMT+5:30 to accept or reject it, otherwise it expires.

Details Info

Requester owner ID
928430096503

Peering connection ID
pcx-0f73c41106c81555b

Status
Pending Acceptance by 928430096503

Expiration time
Thursday, 23 April 2026 at 15:34:37 GMT+5:30

Accept VPC peering connection request Info

Are you sure you want to accept this VPC peering connection request? (pcx-0f73c41106c81555b)

Requester VPC vpc-01e8ea44337907063	Accepter VPC vpc-040e3fcdca0af41a / Vpc1	Requester CIDRs 10.0.0.0/16
Accepter CIDRs -	Requester Region N. Virginia (us-east-1)	Accepter Region Ohio (us-east-2)
Requester owner ID 928430096503(This account)	Accepter owner ID 928430096503(This account)	

Cancel Accept request

```
[root@ip-10-0-1-36 ~]# ping 3.14.252.6
PING 3.14.252.6 (3.14.252.6) 56(84) bytes of data.
64 bytes from 3.14.252.6: icmp_seq=1 ttl=121 time=11.2 ms
64 bytes from 3.14.252.6: icmp_seq=2 ttl=121 time=11.2 ms
64 bytes from 3.14.252.6: icmp_seq=3 ttl=121 time=11.2 ms
64 bytes from 3.14.252.6: icmp_seq=4 ttl=121 time=11.2 ms
64 bytes from 3.14.252.6: icmp_seq=5 ttl=121 time=11.2 ms
64 bytes from 3.14.252.6: icmp_seq=6 ttl=121 time=11.2 ms
64 bytes from 3.14.252.6: icmp_seq=7 ttl=121 time=11.2 ms
64 bytes from 3.14.252.6: icmp_seq=8 ttl=121 time=11.2 ms
64 bytes from 3.14.252.6: icmp_seq=9 ttl=121 time=11.2 ms
^C
--- 3.14.252.6 ping statistics ---
9 packets transmitted, 9 received, 0% packet loss, time 8011ms
rtt min/avg/max/mdev = 11.172/11.187/11.202/0.009 ms
[root@ip-10-0-1-36 ~]#
```

3. Enable VPC peering for cross-account (you can collaborate with your friend to do this task).

- Created VPC peering between **two different AWS accounts**.
- Shared **Account ID and VPC ID** with the acceptor account.
- Accepted the peering request from the second account.
- Updated route tables in both accounts for bidirectional communication.
- Configured security groups to allow traffic between VPC CIDRs.

Select another VPC to peer with

Account

☐ My account

☒ Another account

Account ID

142083399660

Region

☐ This Region (us-east-1)

☒ Another Region

United States (N. California) (us-west-1)

VPC ID (Acceptor)

vpc-08ff784c7704be089

```
[root@ip-10-0-1-36 ~]# ping 54.241.58.54
PING 54.241.58.54 (54.241.58.54) 56(84) bytes of data.
64 bytes from 54.241.58.54: icmp_seq=64 ttl=117 time=63.0 ms
64 bytes from 54.241.58.54: icmp_seq=65 ttl=117 time=62.9 ms
64 bytes from 54.241.58.54: icmp_seq=66 ttl=117 time=62.9 ms
64 bytes from 54.241.58.54: icmp_seq=67 ttl=117 time=62.8 ms
64 bytes from 54.241.58.54: icmp_seq=68 ttl=117 time=62.9 ms
64 bytes from 54.241.58.54: icmp_seq=69 ttl=117 time=62.9 ms
64 bytes from 54.241.58.54: icmp_seq=70 ttl=117 time=62.9 ms
64 bytes from 54.241.58.54: icmp_seq=71 ttl=117 time=62.9 ms
64 bytes from 54.241.58.54: icmp_seq=72 ttl=117 time=62.9 ms
64 bytes from 54.241.58.54: icmp_seq=73 ttl=117 time=63.0 ms
64 bytes from 54.241.58.54: icmp_seq=74 ttl=117 time=62.9 ms
64 bytes from 54.241.58.54: icmp_seq=75 ttl=117 time=62.9 ms
64 bytes from 54.241.58.54: icmp_seq=76 ttl=117 time=62.9 ms
64 bytes from 54.241.58.54: icmp_seq=77 ttl=117 time=62.9 ms
AC
--- 54.241.58.54 ping statistics ---
77 packets transmitted, 14 received, 81.8182% packet loss, time 78568ms
rtt min/avg/max/mdev = 62.843/62.889/62.984/0.044 ms
[root@ip-10-0-1-36 ~]#
```

4. Set up a VPC Transit Gateway.

- Created a Transit Gateway (TGW) to act as a central hub.
- Attached multiple VPCs (VPC, VPC1, VPC2, VPC3) using TGW attachments.

	name	VPC ID	State	Encryption S...	Encryption control ...	Block Public...	
☐	-	vpc-096ec8a65d98c6729	Available	-	-	Off	1
☐	MyVPC	vpc-01e8ea44337907063	Available	-	-	Off	1
☐	VPC1	vpc-068690563bf07f838	Available	-	-	Off	1
☐	VPC2	vpc-0e2dadf1720541ce1	Available	-	-	Off	1
☐	VPC3	vpc-0c1f3e01eeb77d0b7	Available	-	-	Off	1

- Updated route tables in each VPC to route traffic to other VPCs via TGW.

Q 0.0.0.0/0	Internet Gateway	Active	No	CreateRoute	Remove
	Q igw-0604070e37ed1a657				
Q 122.1.0.0/16	Transit Gateway	-	No	CreateRoute	Remove
	Q tgw-06260ab0a3d53ab09				
Q 132.1.0.0/16	Transit Gateway	-	No	CreateRoute	Remove
	Q tgw-06260ab0a3d53ab09				
Q 152.2.0.0/16	Transit Gateway	-	No	CreateRoute	Remove
	Q tgw-06260ab0a3d53ab09				

- Enabled route propagation in TGW route tables for automatic routing.
- Verified connectivity using private IP ping across all VPCs (transitive routing).

Vpc to vpc1

```
[root@ip-10-0-1-36 ~]# ping 44.192.59.183
PING 44.192.59.183 (44.192.59.183) 56(84) bytes of data.
64 bytes from 44.192.59.183: icmp_seq=30 ttl=126 time=0.339 ms
64 bytes from 44.192.59.183: icmp_seq=31 ttl=126 time=0.358 ms
64 bytes from 44.192.59.183: icmp_seq=32 ttl=126 time=0.417 ms
64 bytes from 44.192.59.183: icmp_seq=33 ttl=126 time=0.310 ms
64 bytes from 44.192.59.183: icmp_seq=34 ttl=126 time=0.305 ms
64 bytes from 44.192.59.183: icmp_seq=35 ttl=126 time=0.303 ms
64 bytes from 44.192.59.183: icmp_seq=36 ttl=126 time=0.305 ms
64 bytes from 44.192.59.183: icmp_seq=37 ttl=126 time=0.327 ms
^C
--- 44.192.59.183 ping statistics ---
37 packets transmitted, 8 received, 78.3784% packet loss, time 37441ms
rtt min/avg/max/mdev = 0.303/0.333/0.417/0.036 ms
[root@ip-10-0-1-36 ~]#
```

Vpc to vpc2

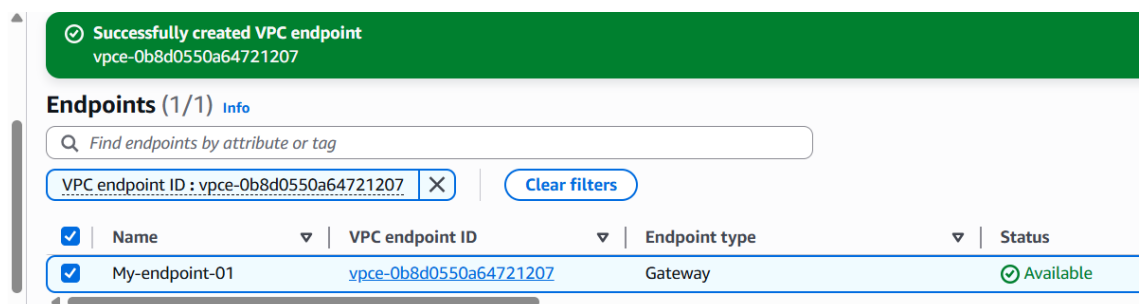
```
root@ip-10-0-1-36:~  
[root@ip-10-0-1-36 ~]# ping 3.221.150.169  
PING 3.221.150.169 (3.221.150.169) 56(84) bytes of data.  
64 bytes from 3.221.150.169: icmp_seq=46 ttl=126 time=0.310 ms  
64 bytes from 3.221.150.169: icmp_seq=47 ttl=126 time=0.319 ms  
64 bytes from 3.221.150.169: icmp_seq=48 ttl=126 time=0.291 ms  
64 bytes from 3.221.150.169: icmp_seq=49 ttl=126 time=0.284 ms  
64 bytes from 3.221.150.169: icmp_seq=50 ttl=126 time=0.313 ms  
64 bytes from 3.221.150.169: icmp_seq=51 ttl=126 time=0.299 ms
```

Vpc to vpc3

```
[root@ip-10-0-1-36 ~]# ping 100.27.37.189  
PING 100.27.37.189 (100.27.37.189) 56(84) bytes of data.  
64 bytes from 100.27.37.189: icmp_seq=1 ttl=126 time=0.401 ms  
64 bytes from 100.27.37.189: icmp_seq=2 ttl=126 time=0.309 ms  
64 bytes from 100.27.37.189: icmp_seq=3 ttl=126 time=0.321 ms  
64 bytes from 100.27.37.189: icmp_seq=4 ttl=126 time=0.309 ms  
64 bytes from 100.27.37.189: icmp_seq=5 ttl=126 time=0.333 ms  
^C  
--- 100.27.37.189 ping statistics ---  
5 packets transmitted, 5 received, 0% packet loss, time 4162ms  
rtt min/avg/max/mdev = 0.309/0.334/0.401/0.034 ms  
[root@ip-10-0-1-36 ~]#
```

5. Set up a VPC Endpoint.

- Created a **Gateway VPC Endpoint for Amazon S3**.



- Selected appropriate VPC and associated route tables.
- Automatically added route:
pl-63a5400a → vpce-0b8d055xxxx07 in route table.

Destination	Target	Status
pl-63a5400a	vpce-0b8d0550a64721207	✓ Active
10.0.0.0/16	local	✓ Active
	Q local	X

- Enabled private access to S3 without Internet Gateway or NAT.
- Create access key in security guidliness

ot user access keys **Retrieve access key** [Info](#)

Access key
If you lose or forget your secret access key, you cannot retrieve it. Instead, create a new access key and make the old key ir

Access key	Secret access key
AKIA5QKV7AB3RRBO2LN3	***** Show

- Verified using aws s3 ls from EC2 instance.

```
[root@ip-10-0-1-36 ~]# aws s3 ls
Unable to locate credentials. You can configure credentials by running "aws login".
[root@ip-10-0-1-36 ~]# aws configure
AWS Access Key ID [None]: AKIA5QKV7AB3RRBO2LN3
AWS Secret Access Key [None]: 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
Default region name [None]: us-east-1
Default output format [None]: json
[root@ip-10-0-1-36 ~]# aws s3 ls
2026-04-15 15:15:35 my-vpc-flowlogs-bucket-123
2026-04-15 16:02:26 my-vpc-flowlogs-bucket-456
[root@ip-10-0-1-36 ~]#
```